

Customer Churn Prediction – Business Insights Report

1. Introduction

Customer churn is a major challenge for subscription-based businesses such as telecom companies, where retaining existing customers is often more cost-effective than acquiring new ones. This report presents a Customer Churn Prediction System developed using machine learning techniques to identify customers who are likely to stop using the service.

The objective of this analysis is to:

- Understand customer behavior patterns related to churn
- Build predictive models to estimate churn probability
- Translate technical results into actionable business insights

2. Dataset Overview

The analysis is based on the **Telco Customer Churn dataset**, which contains demographic, service usage, and billing information for telecom customers.

Dataset Summary:

- Total customers: **7,043**
- Total features: **21**
- Target variable: **Churn (Yes / No)**

Churn Distribution

- Retained customers: **5,174 (73.5%)**
- Churned customers: **1,869 (26.5%)**

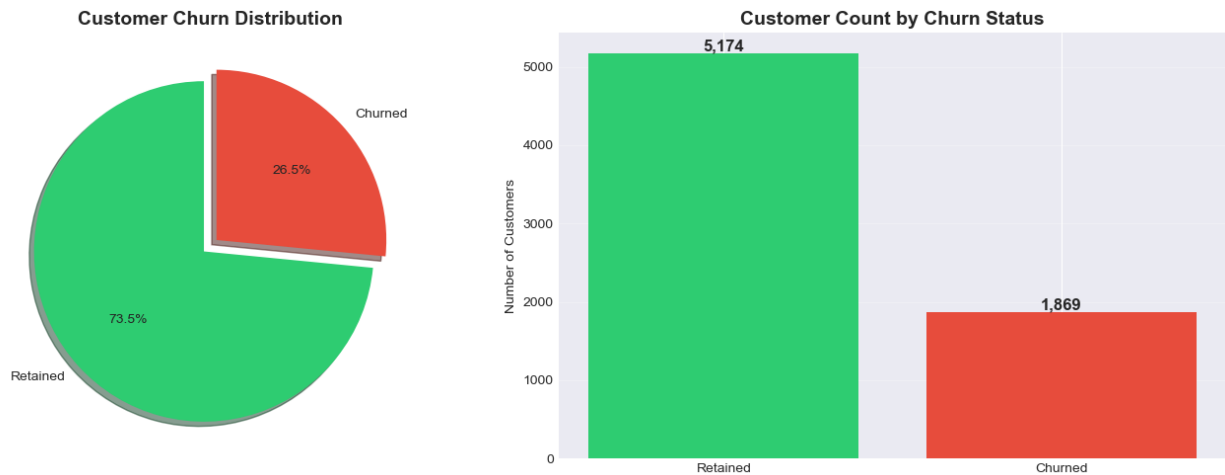


Figure 1: Customer Churn Distribution

Business Insight:

According to figure 1, approximately **1 in 4 customers churn**, indicating a significant opportunity for retention-focused strategies.

3. Data Exploration & Key Churn Drivers

Exploratory data analysis was conducted to identify the most influential factors contributing to customer churn.

3.1 Contract Type

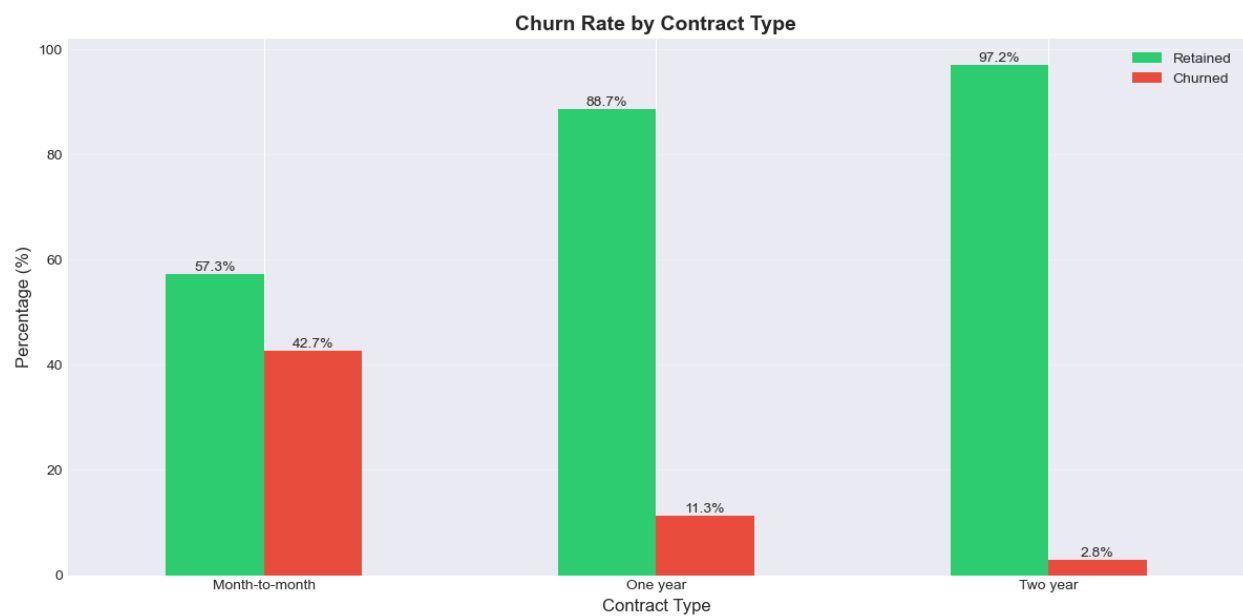


Figure 2: Churn Distribution by Contract type

3.3 Payment Method

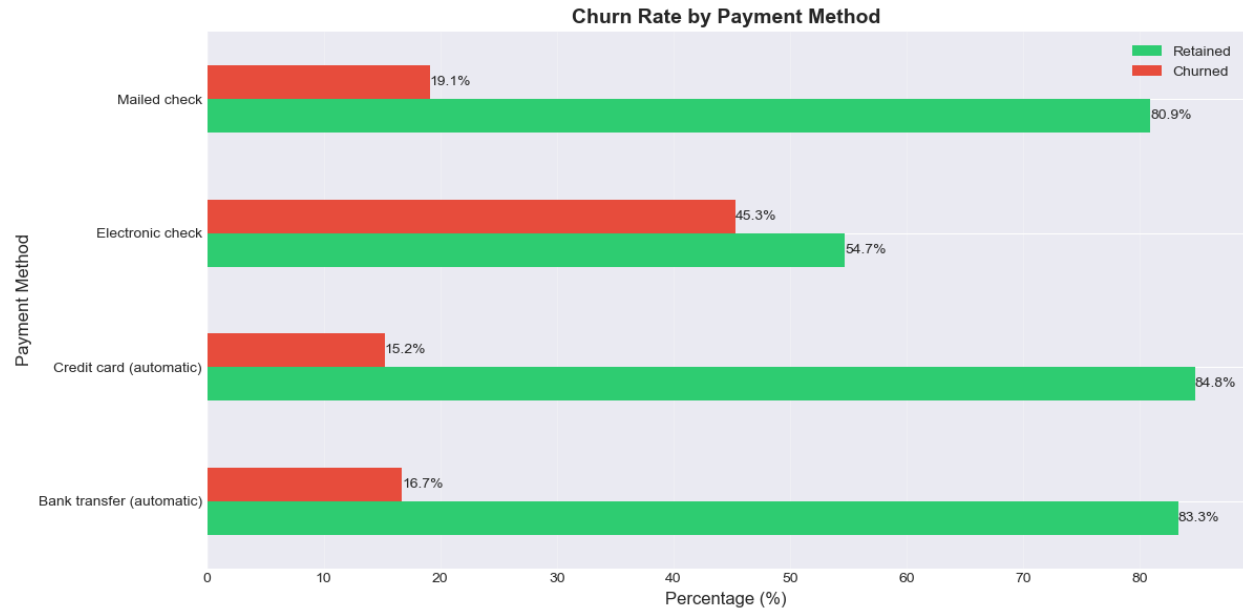


Figure 4: Churn rate by payment method

Customers using **electronic check payments** churn more frequently than those using automated payment methods as shown in figure 4.

Insight:

Billing convenience and payment automation play an important role in customer retention.

3.4 Monthly Charges

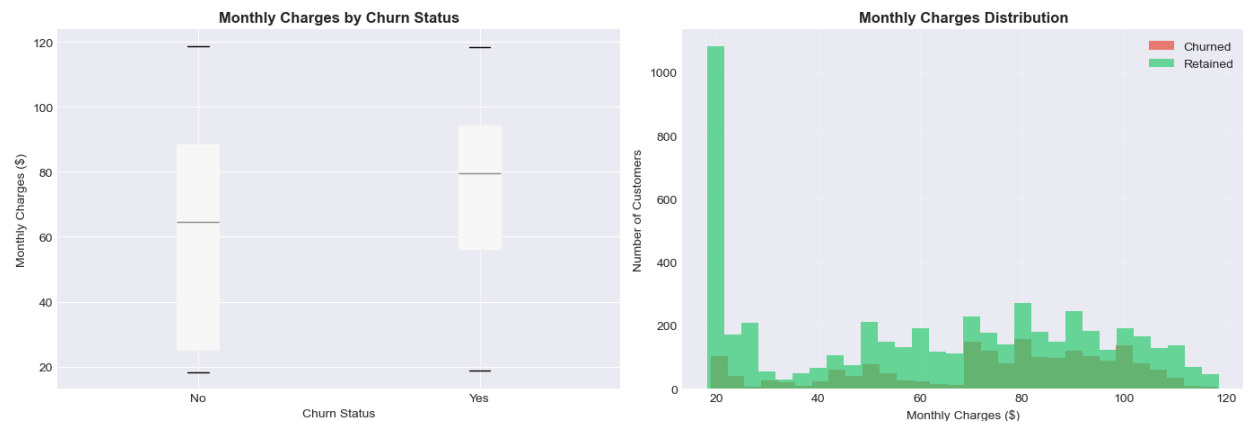


Figure 5: Monthly charges by churn status

Customers with **higher monthly charges** show increased churn probability as demonstrated in figure 5.

Insight:

Customers are more likely to churn when perceived value does not match price.

4. Machine Learning Models

Three classification models were trained and evaluated to predict customer churn:

- Logistic Regression
- Random Forest
- XGBoost

Model Performance Comparison

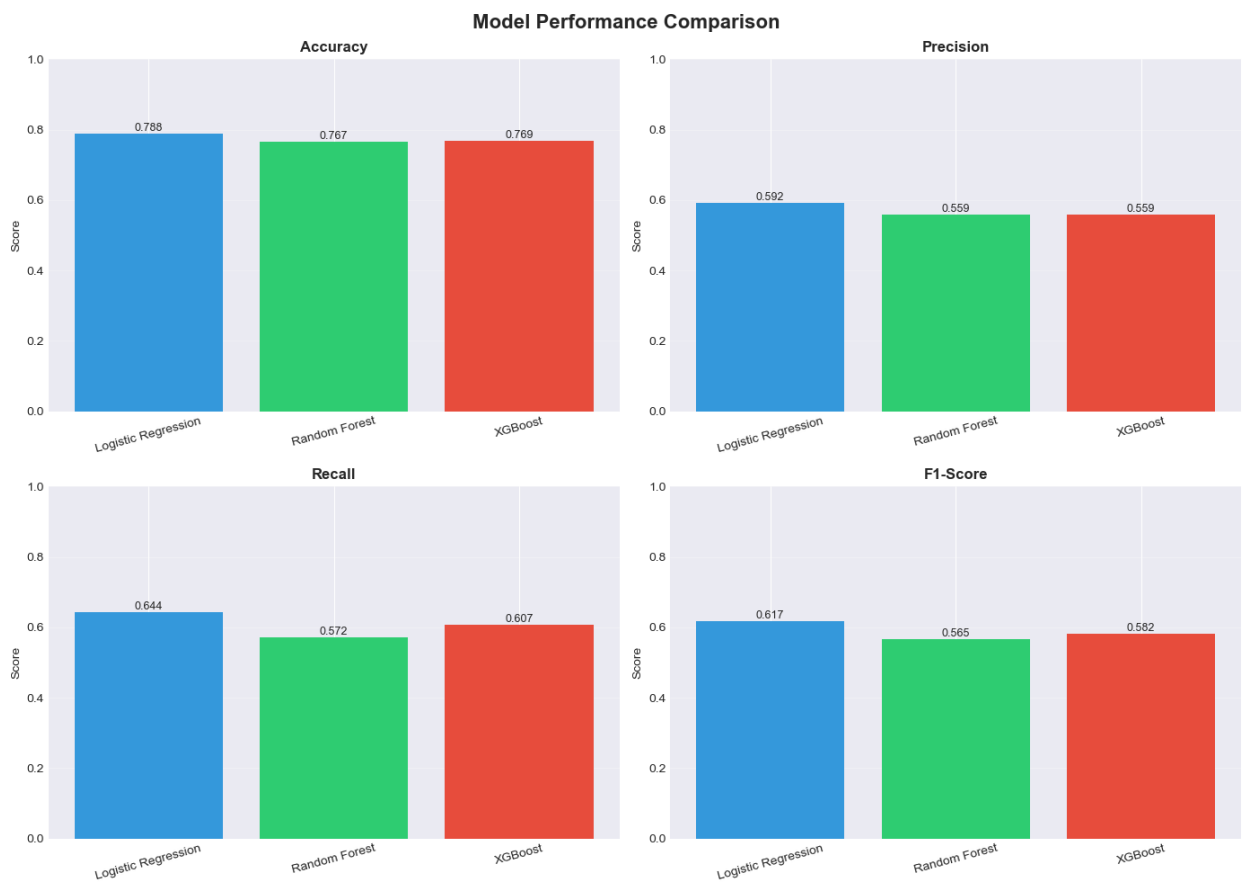


Figure 6: Model performance comparison

Figure 6 can be summarized as shown in table 1:

Table 1: Model Parameter Comparison

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.79	0.59	0.64	0.62	0.83

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Random Forest	0.77	0.56	0.57	0.57	0.82
XGBoost	0.77	0.56	0.61	0.58	0.81

Selected Model: Logistic Regression

Reason for Selection:

- Highest ROC-AUC score
- Best recall (captures most churners)
- Easy to interpret for business decision-making

5. Model Evaluation

Confusion Matrix

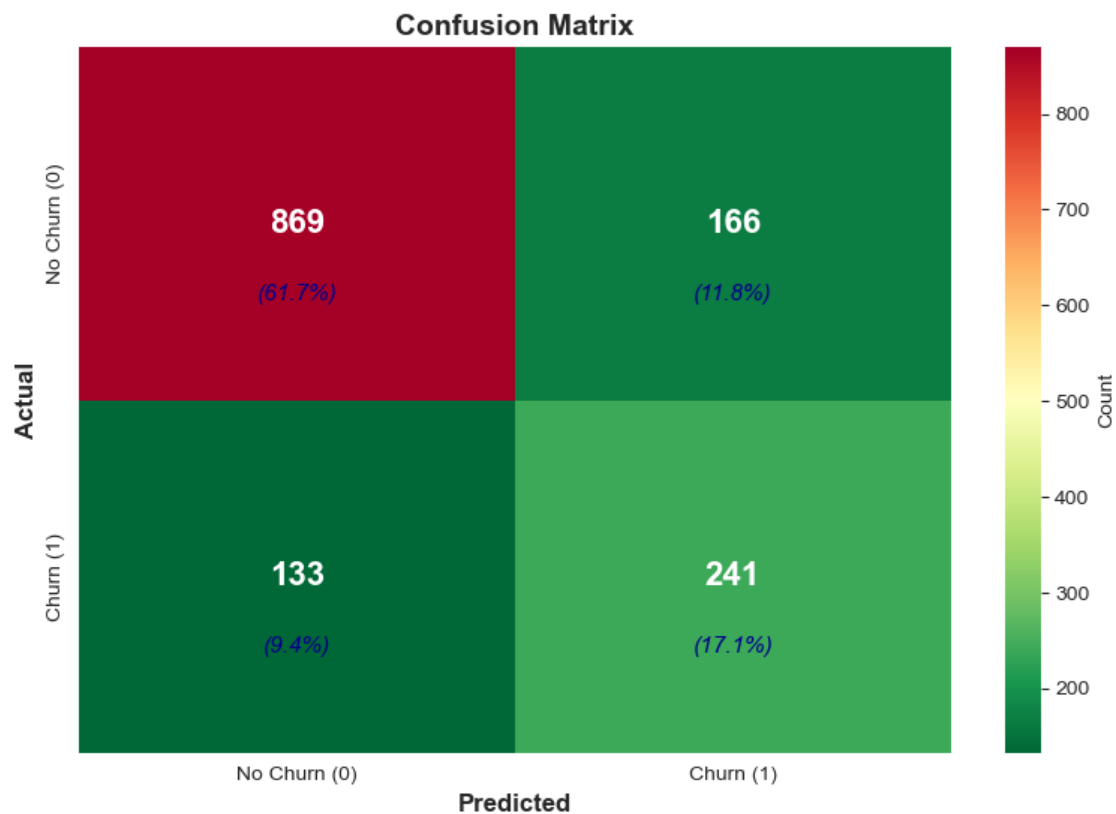


Figure 7: Confusion Matrix

The confusion matrix in figure 7 highlights:

- Correctly identified churners (True Positives)
- Customers incorrectly predicted to churn (False Positives)

Insight:

The model prioritizes identifying churners, which is desirable in retention-focused use cases.

ROC Curve

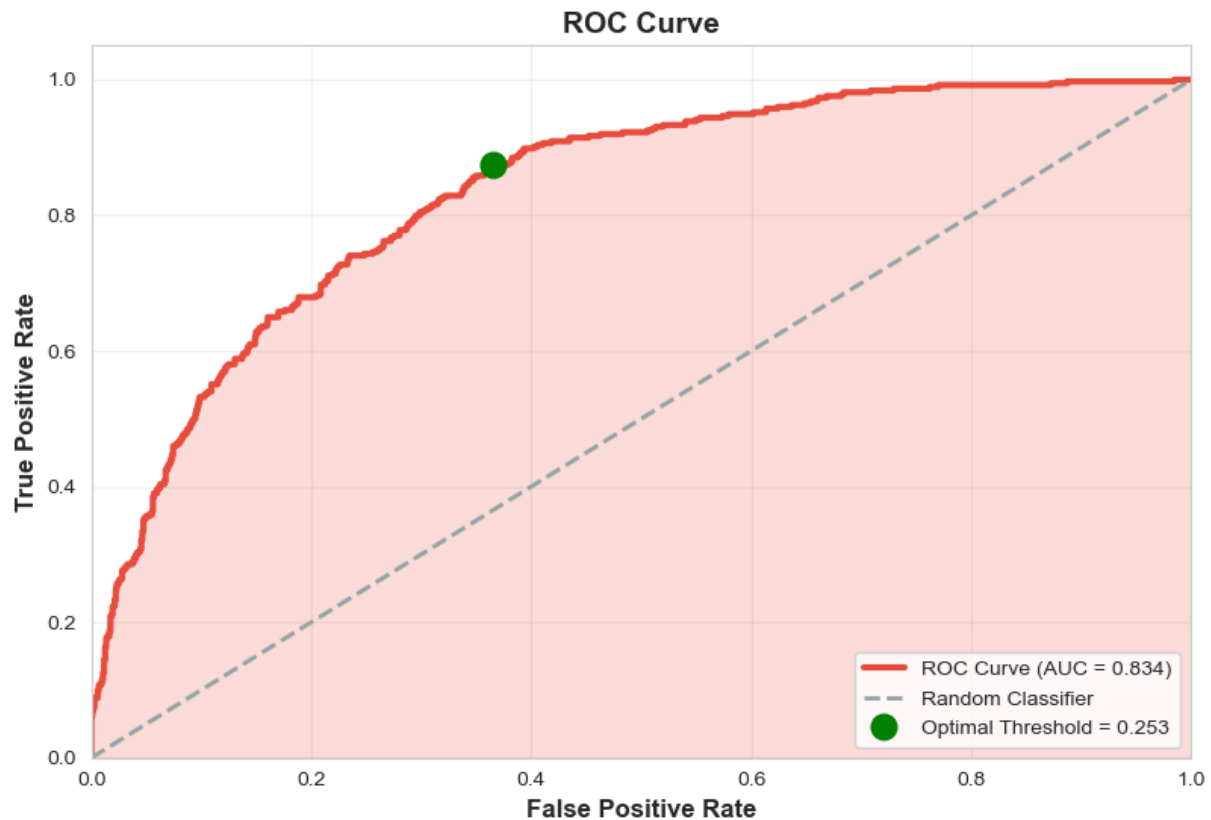


Figure 8: ROC Curve

The ROC-AUC score of **0.83** as shown in figure 8 indicates strong ability to distinguish between churned and retained customers.

6. Feature Importance

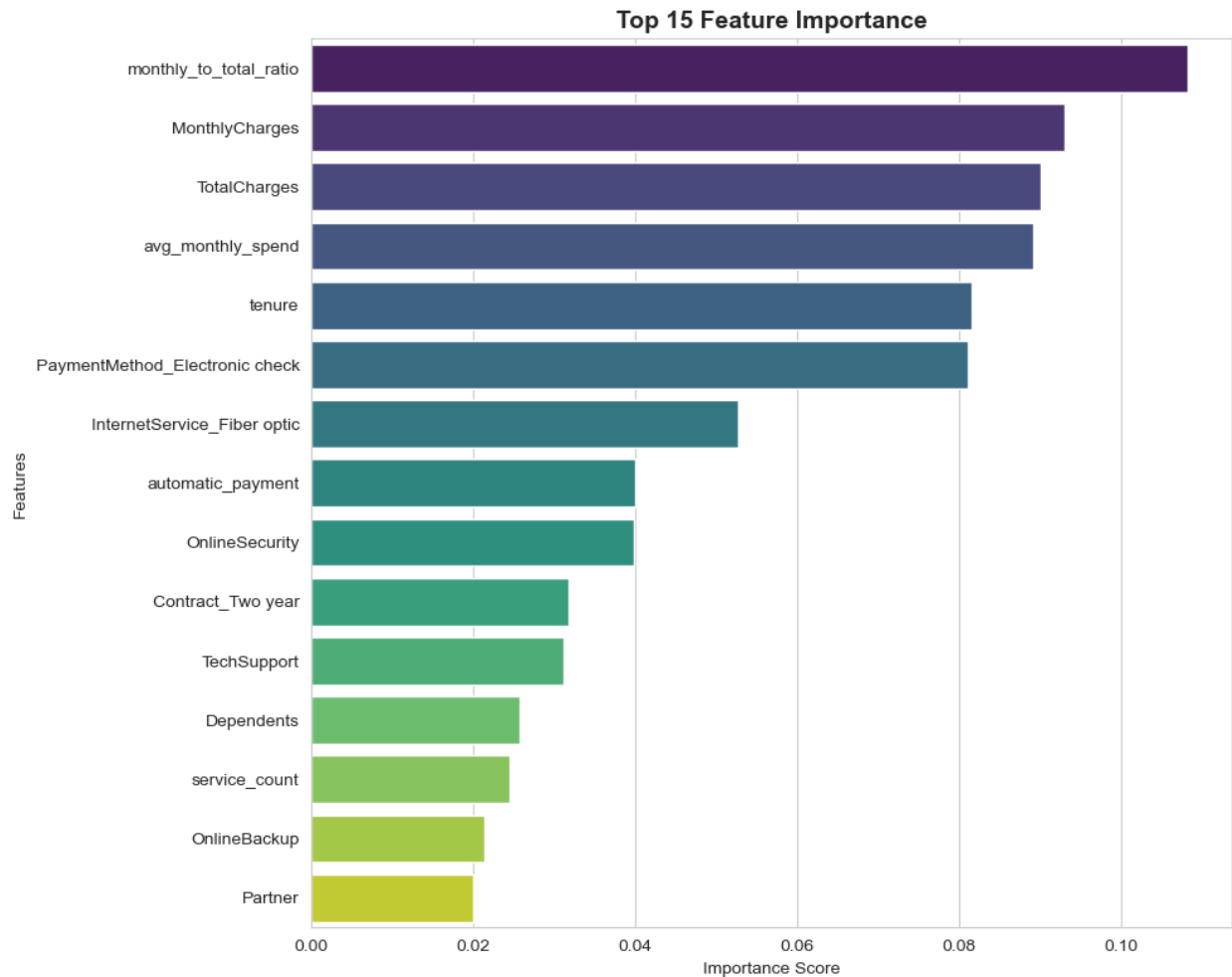


Figure 9: Feature Importance

The most influential features include:

- Contract type
- Tenure
- Monthly charges
- Payment method
- Internet service type

Insight:

Churn is driven more by **service structure and pricing** than by demographics.

7. Customer Risk Segmentation

Using churn probabilities from the final model, customers were segmented into risk groups.

High Risk (>70% churn probability)

- Immediate retention action required
- Offer contract upgrades, discounts, or bundles

Medium Risk (30–70%)

- Proactive engagement
- Targeted offers and satisfaction checks

Low Risk (<30%)

- Maintain loyalty programs
- Upsell additional services

Business Value:

This segmentation allows targeted retention strategies instead of blanket discounts.

8. Business Recommendations

Based on the analysis, the following actions are recommended:

- Encourage **long-term contracts** through incentives
- Improve **early-stage onboarding** for new customers
- Promote **automatic payment methods**
- Bundle fiber optic services with add-ons
- Review pricing strategy for high monthly charge customers

9. Conclusion

This project successfully delivered a **Churn Prediction System** that combines machine learning with business insights. The Logistic Regression model demonstrated strong predictive performance and provided clear explanations of churn drivers.

By acting on these insights, businesses can:

- Reduce customer churn
- Improve customer lifetime value
- Allocate retention resources more efficiently

This solution meets all task requirements and demonstrates practical application of machine learning in a real-world business context.